NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/48A(USAF)
28 March 1991

SUPERSEDING
MIL-R-6106/48(USAF)
22 March 1989

PERFORMANCE SPECIFICATION SHEET
INCH-POUND
RELAYS, ELECTROMAGNETIC, TYPE I, HERMETICALLY SEALED, 100 AMPERES, SPST (N.O.) WITH SPDT (FORM Z) 5 AMPERES AUXILIARY CONTACTS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

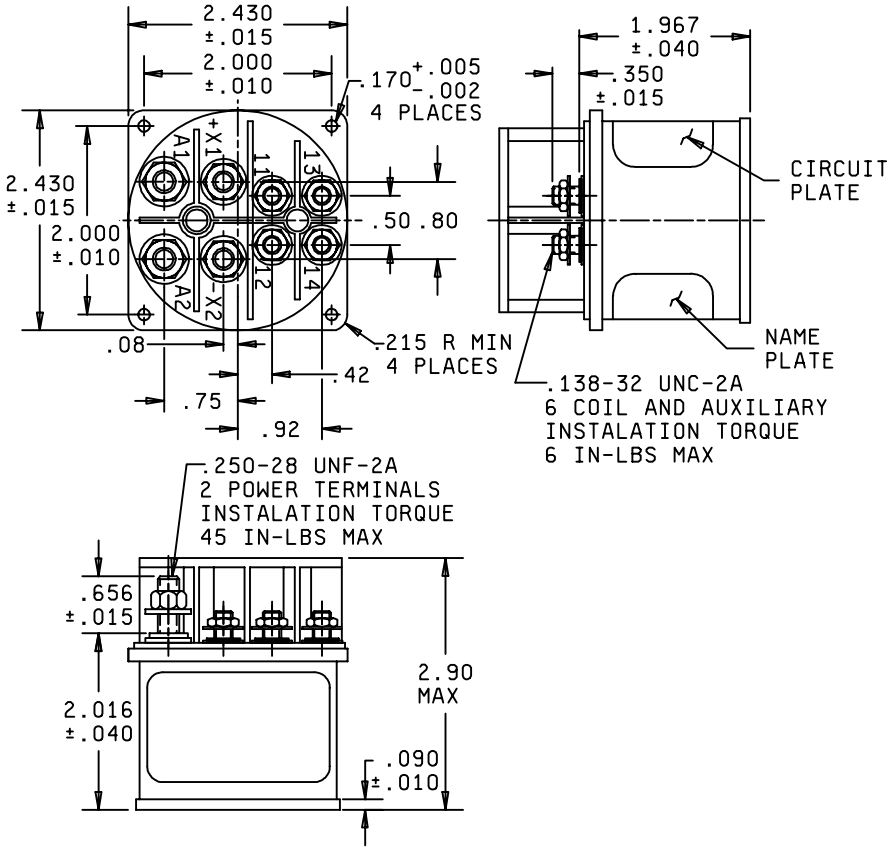
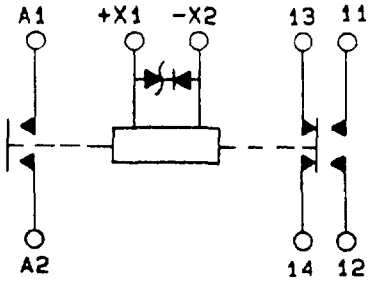


FIGURE 1. Relay, outline drawing.

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CIRCUIT DIAGRAM

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.138	3.51	.800	20.32
005	0.13	.170	4 32	.915	23.24
.010	0.25	.250	6.35	1.967	49.96
.015	0.38	.350	8.89	2.000	50.80
.040	1.02	.415	10.54	2.016	51.12
.085	2.16	.656	16.66	2.430	61.72
.090	2.29	.750	19.05	2.90	73.7

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.76 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Terminal number need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.

FIGURE 1. Relay, outline drawing - Continued.

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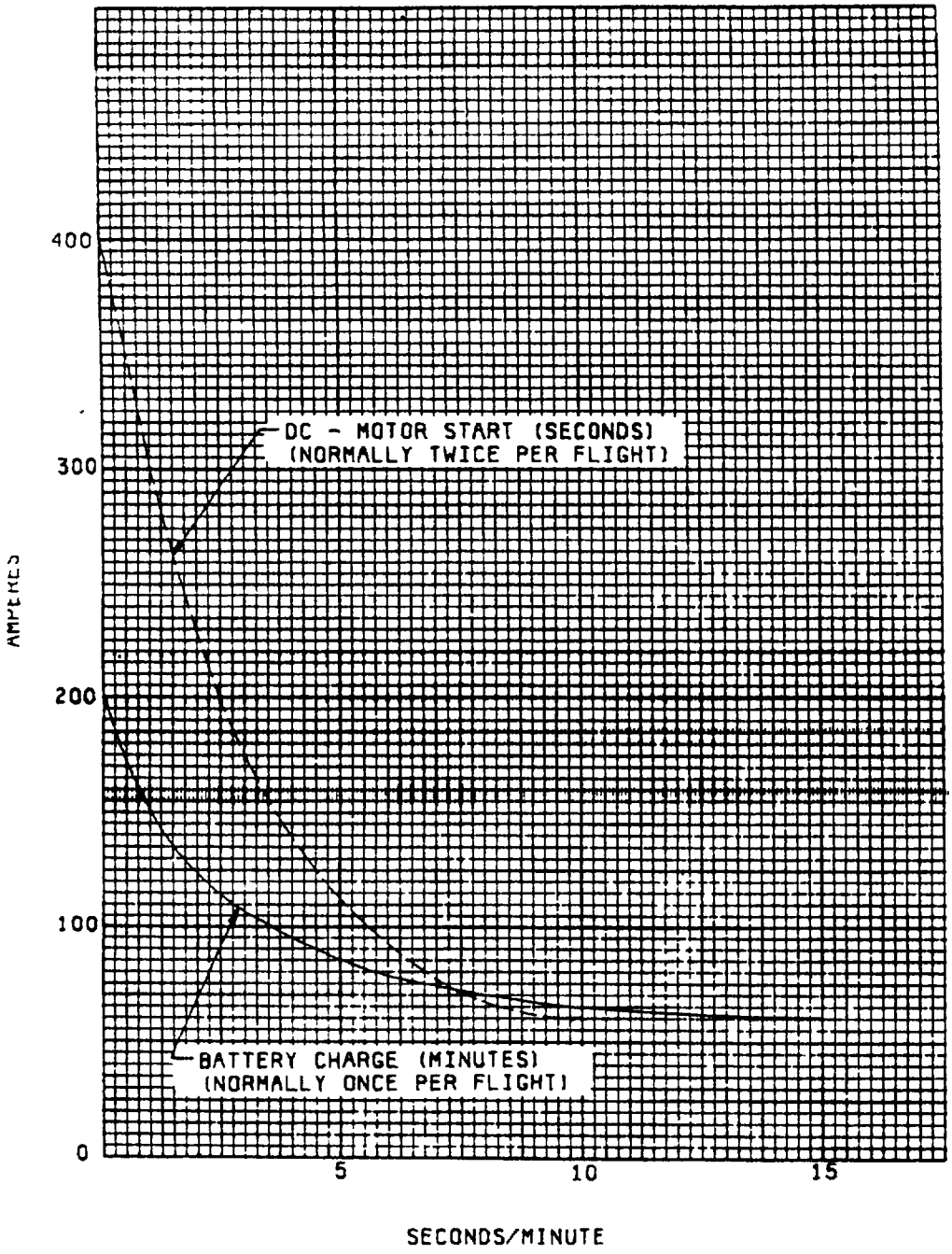


FIGURE 2. Overload.

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REQUIREMENTS

Coil data

Duty cycle Continuous

Coil voltage 28 V dc, maximum 30 V dc

Pickup voltage, maximum 18 V dc, over the temperature range, 15 V dc at +25°C

Pickup voltage, high temperature test 21 V dc maximum.

Pickup voltage, continuous current test 22.5 V dc maximum.

Hold voltage 7.0 V dc, over the temperature range.

Dropout voltage 1.5 V dc, over the temperature range.

Coil resistance: 100 ohms minimum at ±25°C.

Coil transient voltage: 42 V peak inverse voltage

Contact data

Contact voltage drop: 100 mV maximum initial, 150 mV maximum after life test.

Operate time 30 milliseconds maximum at ±25°C with rated coil voltage.

Release time 20 milliseconds maximum at ±25°C from rated coil voltage.

Contact bounce 3 milliseconds maximum (main and auxilary contacts).

Main contacts:

Voltage ratings: 28 V dc or 115 V ac, 400 Hz.

Current rating:

Resistive: 100 amperes.

Inductive 100 amperes (10,000 life cycles).

Motor 100 amperes.

Lamp 50 amperes (25,000 life cycles).

Intermediate current: 10 amperes.

Overload current: 800 amperes and as shown in figure 2.

Rupture current: 1,000 amperes.

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Auxiliary contacts

Voltage rating 28 V dc or 115 V ac, 400 Hz.

Resistive 5 amperes.

Inductive 5 amperes (10,000 life cycles).

Lamp 1 ampere (25,000 life cycles).

Mechanical life at reduced current 0.5 ampere (100,000 life cycles)

Intermediate current 2 milliamperes at 28 V dc. 50 ohms maximum contact resistance 1/

Contact voltage drop 5 millivolts maximum at 6 V dc and 100 mA. 1/

Physical data

Strength of terminals. Soldered terminals shall be tested in accordance with the following

	Torque test in-lb		Force test lb		Installation torque in-lb
Terminal size	Room temp.	180°C	Room temp.	180°C	
6 - 32	10	4	30	12	6
1/4 - 28	75	30	50	20	45

Weight 11.0 ounces (312 grams) maximum.

Finish: Optional (painted or plated).

Environmental data

Ambient temperature range. -55°C to +125°C.

Altitude· 80,000 feet maximum.

Shock: 50 g's (half sine wave).

Duration: 6-9 milliseconds.

Maximum duration contact opening· 2 milliseconds.

Vibration random 15 minutes each plane, 0.0375 G/Hz for 10-125 Hz, 4 DB/octave increase 125-250 Hz, 0.1 G/Hz for 250-1000 Hz, 3 DB/octave decrease 1000-2000 Hz.

Vibration sinusoidal· 55-2000 Hz, 10 g's.

At 5 - 10 Hz 0.08 da.

At 10 - 55 Hz 0.05 da.

Salt spray Applicable.

1/ CAUTION Auxiliaries tested or used above 100 milliamperes at 6 V dc maximum or peak ac open circuit, will negate the intermediate current requirement.

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Electrical data

Life:

Electrical 50,000 cycles minimum at rated load (unless otherwise specified)

Mechanical 100,000 cycles.

Insulation resistance, initial 100 megohms.

After life or environmental tests 50 megohms.

Dielectric withstanding voltage (sea level)

	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	1250	1000
Auxiliary contacts	1250	1000
All other points	1800	1350

Dielectric withstanding voltage (altitude):

	80,000 ft V rms (60 Hz)
Coil to case	500
Auxiliary contacts	500
All other points	700

Part or Identifying Number (PIN). M6106/48-001.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian.
Air Force - 85

Review activity.
Air Force - 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F720)

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